Technical Appendix

Multi-agent Based Measurement of Organizational Culture

Sections A to D specify the instrument and its validation protocol; Section E states the reproducibility commitments; Section F covers ethics and data access; Section G lists the contents of the accompanying anonymized repository; Section H reports the first prototype runs descriptively. The prompts in Section B are those of the full eleven-role design; the five-role prototype in the repository implements the Reader, Evidence skeptic, Judge, Interpretive critic and Adjudicator roles with these prompts and runs the six expertise agents as an optional stage; earlier prompt versions are kept in the repository for comparison.

A. Corpus, pre-processing and masking

A.1 Corpus and provenance

Transcripts come from a standard earnings-call transcript corpus covering listed firms over multiple years; the provider, universe, period and observed coverage (firm-quarters with a transcript as a share of the universe) are reported in the repository documentation and with the first release of the instrument. The validation set (Section D.4) will contain approximately 500 passages. Where the provider's terms prevent redistribution of full transcripts, the validation passages, their masked windows and all agent logs are released in the form the terms permit, and a replication subset is built from publicly filed transcripts (for example, transcripts filed as exhibits with the regulator or posted on investor-relations pages) so that every reported number can be regenerated.

A.2 Segmentation into passage windows

Transcripts are split into speaker-separated turns. Prepared remarks are segmented into short passages with a bounded context window of a fixed number of sentences on either side; both lengths are parameters recorded in the repository configuration and held fixed across arms. Analyst questions are kept attached to the answer that follows them so that qualifications offered in reply are not separated from the claim they qualify. Duplicate descriptions of the same episode across quarters are linked by lexical similarity within firm and recorded once as an episode with multiple mentions; recurring practices mentioned in several quarters are retained in each quarter and linked. Operator notices, safe-harbor statements and legal disclaimers are removed by pattern rules that are released with the code.

A.3 Masking

Firm, person, product and location names are replaced with typed, numbered placeholders that are stable within a transcript ([COMPANY_001], [PERSON_002], [PRODUCT_003], [LOCATION_004]). Named-entity detection uses a standard named-entity recognizer with a manual check on the validation set. Dates and monetary amounts are retained, because reported actions often carry quantities that the Reader must record; the re-identification probe below quantifies the leakage this permits, and if the probe shows substantial re-identification, dates are replaced by relative markers (THIS_YEAR, LAST_QUARTER), monetary amounts are bucketed into magnitude bins (AMOUNT_10M_100M), and the probe is repeated.

Constructed example (no real firm).

```
Original: "Thanks, Maria, and good morning. Northgate Industrial had a solid year. We sold
three non-core plants, cut net debt by roughly a third, and opened two sites in Ohio.
```

```
Every plant manager now reviews safety incidents with their team each Monday."

Masked:   "Thanks, [PERSON_002], and good morning. [COMPANY_001] had a solid year. We sold
three non-core plants, cut net debt by roughly a third, and opened two sites in
[LOCATION_001]. Every plant manager now reviews safety incidents with their team
each Monday."
```

Re-identification probe. For every passage in the validation set, the pinned model is asked, in a separate call with no other context, to name the firm and the quarter from the masked window. The share of correct firm identifications is reported as a measure of residual leakage, under each masking scheme used.

A.4 Screening for scale

The pilot and the validation set use unscreened passages, so no evidence is lost to a filter. At scale, a low-cost relevance screen (a small classifier or a single short model call) removes passages with no reported conduct. Recall of the screen is estimated on the validation sample, which is drawn before screening and carries adjudicated reference labels: recall is the share of reference passages containing codable conduct that the screen retains. Auditing screened-out passages alone would estimate only the share of exclusions that contain relevant content, not recall. The screen is accepted for a reported run if the point estimate of recall is at least 0.95 and the lower one-sided 95 percent confidence bound is at least 0.90; precision is reported alongside. The estimate and the screened-out share are published with every screened run.

B. Roles, information boundaries and prompts

Table B1. What each role sees, does not see, and produces.

Role	Sees	Does not see	Produces	Budget
Reader	Masked passage and context window	Codebook, culture labels	Evidence record; declaration ledger	1 record, 1 reply
Evidence skeptic	Masked passage, context window, Reader record	Codebook, culture labels, downstream outputs	Typed flags with targeted questions	1 critique
Six expertise agents	Interpretation view of the record (Section B.0), one category definition	Passage, declaration ledger, other experts, Judge	Category assessment (JSON)	1 call each
Judge	Interpretation view, codebook, six assessments	Passage, declaration ledger, identifiers	Claims; reply	2 statements
Interpretive critic	Same as Judge, plus Judge claims (and reply in arm F)	Passage, declaration ledger, identifiers	Challenge; reassessment	2 statements
Adjudicator	Interpretation view, codebook, four Judge–critic statements	Passage, identifiers, outcome data, declaration ledger	Final measurement record; process record	1 decision

B.0 The interpretation-stage input view

The pipeline constructs a separate object, `record_interp`, for every call in the interpretation stage. It is derived from the revised record by removing the passage, the context window, the declaration ledger and the character spans of excerpts, and by keeping (i) the conduct items with their fields and verified excerpt text (evidence IDs E1, E2, ...), and (ii) a sanitized list of the unresolved source objections: flag type, affected evidence IDs where applicable, a one-line description with no passage text beyond the verified excerpts, the Reader's marking of whether the passage states the missing information (`source_states`: absent, unclear or present), and a passage-level `completeness_uncertain` indicator that is true whenever a MISSING_ITEM flag remains unresolved. Without (ii) a record could look complete to the interpretation agents after the source stage had identified an unresolved omission, and the adjudicator could not tell whether a gap was inherent in the source or introduced by the Reader. Passing the complete record with an instruction to ignore a field would not implement the boundary; the restriction is enforced in the inputs, and the logged inputs of every interpretation call can be checked for it. "Verified excerpts" are the short supporting excerpts recorded by the Reader and confirmed by the skeptic; exact string matching checks provenance, and whether the excerpt supports the claim is the skeptic's task.

B.1 Reader prompt

```
You are the Reader in a measurement system for earnings-call transcripts.
You receive one masked passage and its context window. Nothing else.

TASK: build an evidence record of reported conduct. Do not classify culture.

RULES
1. Record only what the passage states. Do not infer motives, rules or norms.
2. For each item of reported conduct give: actor (placeholder or role); conduct;
   timing; status (completed | ongoing | planned); content_type (expectation |
   rule | routine | episode); normative direction if stated (required |
   encouraged | permitted | discouraged | sanctioned | unclear); origin if
   stated (internal | external | unstated); scope as three fields: scope_type
   (organization-wide | unit or function | team or role | unspecified),
   scope_target (the unit, team or role named, verbatim or placeholder) and
   scope_extent (one | some | all | named | unspecified); qualifications
   stated in the passage; and a verbatim supporting excerpt of at most 25 words
   with an evidence ID (E1, E2, ...).
3. content_type: "routine" only when the passage describes the conduct as
   standard practice for a defined set of actors ("every", "each", "all",
   "monthly"); otherwise a reported past action is an "episode".
4. Statements of values or intent with no reported conduct ("we champion
   innovation") go to the DECLARATION LEDGER, not the record.
5. If the passage is ambiguous, record the ambiguity as a qualification.
6. Output valid JSON following the record schema.

WHEN REPLYING TO THE SKEPTIC: for every flag, amend or defend the record citing
the passage, and set source_states for the flagged information to "absent" if
the passage does not state it, "unclear" if it is ambiguous, or "present" if
you can now cite it. Unresolved flags keep their source_states value.
```

B.2 Evidence skeptic prompt

```
You are the Evidence Skeptic. You receive the masked passage, its context
window and the Reader's record. You do not know how the record will be used.
```

```
TASK: find every way in which the record misrepresents OR under-represents
the passage. Check, in order:
  (a) UNSUPPORTED_CLAIM: paraphrase that goes beyond the text;
  (b) OMITTED_QUALIFIER: conditions, hedges or scope limits left out;
  (c) IGNORED_COUNTEREVIDENCE: statements in the window that cut against an item;
  (d) SCOPE_OR_STATUS_ERROR: wrong timing, status, scope fields, direction,
      origin or content_type;
  (e) WEAK_EXCERPT: an excerpt that does not support the item it is attached to;
  (f) MISSING_ITEM: reported conduct in the passage that appears nowhere in the
      record. Read every sentence of the passage and list each independent
      conduct item that is absent, citing the sentence number.

For each flag give its type, the target item (or "none" for MISSING_ITEM), the
sentence(s) concerned, and one targeted question the Reader can answer by
re-reading the passage. Do not propose culture labels. Do not rewrite the
record yourself. If the record is complete and fully supported, say so.
Output valid JSON (critique schema).
```

The Reader receives the critique and replies once. The reply may amend the record (adding, correcting or removing items) or defend it, citing the passage, and marks `source_states` for each flag. Flags the Reader does not resolve are carried into the revised record and, in sanitized form, into the interpretation view. Flag types are logged so that missing-item repairs can be reported separately from unsupported-claim repairs. In arm C (Section D.2) the critique is generated but withheld, and the Reader instead re-reads the passage with a generic instruction to check its own record for the same six error types without being told which apply.

B.3 Expertise agent prompt and category definitions

Six copies of one template run in parallel, each instantiated with a single category definition and its indicator list. The template analyzes the interpretation view and never sees the passage; the predecessor pilot, by contrast, applied category-specialist prompts to raw transcript components, which is the design this template replaces.

```
You are the {CATEGORY} specialist in a multi-agent measurement system.

INPUT: the interpretation view of a verified evidence record from one
earnings-call passage: conduct items (actor, conduct, timing, status,
content_type, direction, origin, scope_type, scope_target, scope_extent,
qualifications, verified excerpt E1, E2, ...) and any unresolved source
objections with their source_states. You do NOT receive the transcript. Treat
the view as the only evidence source.

GLOBAL RULES
1. Do not use outside knowledge about any firm, industry or event.
2. Do not infer conduct that the view does not contain.
3. Do not request more context.
4. Prefer "insufficient evidence" to guessing.
5. Reported normative content means conduct that is required, encouraged,
   permitted, discouraged or sanctioned inside the organization, or a routine
   described as standard practice for a defined set of actors. An episode
   (a one-off reported action) is evidence of at most WEAK relevance on its
   own. Strategy, leadership decisions, financial results and market
   conditions count only when the view links them to internal conduct.
6. Plans and declarations never establish implemented practice.
7. A requirement imposed from outside (regulation, customer) still counts as
```

```
   reported normative content; note its origin, do not exclude it.
8. If completeness_uncertain is true, say so in unresolved; do not treat the
   view as complete.
9. Cite evidence by ID only. Output valid JSON (assessment schema).

CATEGORY DEFINITION: {DEFINITION}
INDICATORS: {INDICATORS}
NON-QUALIFYING SIGNALS: {NON_QUALIFYING}

TASK: return one entry per distinct expectation, rule or routine in the view
that concerns {CATEGORY} (a passage may contain several), each with:
- evidence_ids; reasons_for and reasons_against (one sentence each, by ID);
- relevance: none | weak | moderate | strong, "strong" only if reasons_for
  clearly outweigh reasons_against;
- direction (required | encouraged | permitted | discouraged | sanctioned |
  unclear) and origin, copied from the view;
- scope_type, scope_target, scope_extent, copied from the view;
- unresolved: qualifications or source objections that limit the entry.
If nothing in the view concerns {CATEGORY}, return relevance none.
```

Codebook provenance. The published article of Li et al. (2026, Section 1.2) defines six culture types: collaboration and people-focused, customer-oriented, innovation and adaptability, integrity and risk management, performance-oriented, and miscellaneous. The five substantive categories below are those types; their topic lists follow the definitions given in that article and its Internet Appendix. Two things are our adaptation and are stated as such. First, the coded unit is reported normative content about these topics (expectations, rules and routines), so each definition is introduced as "reported expectations, rules or routines concerning", not as "a culture focusing on". Second, the residual category is redefined as **Other reported norms**: reported normative content that fits none of the five. Li et al.'s Miscellaneous type covers generic culture talk; in our design generic declarations without reported conduct never reach the interpretation stage, so they are counted in the declaration ledger and reported descriptively. The single-pass baseline (Section D.5) uses this same adapted codebook; the faithful implementation of Li et al.'s procedure uses their original definitions. The indicator and non-qualifying lists are the authors' operationalization and are part of the preregistered codebook.

Table B2. Category definitions, indicators and non-qualifying signals supplied to each expertise agent.

Category	Definition	Indicators	Non-qualifying signals
Collaboration and people-focused	Reported expectations, rules or routines concerning: collaboration, cooperation, teamwork, supportive environment, low levels of conflict, community, communication within the organization, employee well-being, employee equity sharing and compensation, diversity, inclusion, empowerment, or talent development.	collaboration, teamwork, cross-functional work, supportive environment, psychological safety, internal communication, information sharing, employee well-being, work-life balance, morale, compensation fairness, profit sharing, diversity and inclusion, empowerment, autonomy, voice, talent development, mentoring, learning culture, retention, engagement.	customers, markets, financial metrics, risk or compliance, or efficiency, unless linked to how people or teams operate internally.

Category	Definition	Indicators	Non-qualifying signals
Customer-oriented	Reported expectations, rules or routines concerning: sales orientation, customer service, listening to the customer, customer retention, customer experience, customer satisfaction, user experience, client service, being brand-driven, quality of product, quality of service, quality of solution, or taking pride in service.	customer service, listening to customers, customer feedback loops, retention, customer experience, satisfaction scores used internally, service quality standards, product quality standards, pride in service, brand promise translated into conduct.	customer demand or market growth reported as outcomes, sales figures, pricing, or competitive positioning without reference to how employees serve customers.
Innovation and adaptability	Reported expectations, rules or routines concerning: innovation, creativity, technology adoption, entrepreneurship, adaptability, transformation, flexibility, agility, willingness to experiment, moving beyond tradition, disruption, being fast-moving, quick to take advantage of opportunities, resilience to change, or taking initiative.	experimentation, pilots, learning from failure, technology adoption as practice, agility in reorganizing work, initiative taking, tolerance for risk in new ideas, speed of decision making, continuous improvement routines, intrapreneurship.	product launches, R&D spending, or "transformation" programs reported as outcomes or plans without reference to how employees are expected to behave.
Integrity and risk management	Reported expectations, rules or routines concerning: integrity, high ethical standards, honesty, transparency, accountability, doing the right thing, fair practices, being trustworthy, risk management, risk control, compliance, discipline, financial prudence, safety, or internal control.	ethics standards, honesty, transparency with stakeholders, accountability routines, fair dealing, compliance practices (internal or externally required; record origin), risk review routines, safety routines, internal control practices, disciplined capital allocation as a conduct expectation.	regulatory events, litigation, impairments, leverage ratios or capital returns reported as outcomes without reference to internal conduct expectations.
Performance-oriented	Reported expectations, rules or routines concerning: high expectations for performance, achievement, competitiveness, results, hard work, efficiency, productivity, consistency in executing tasks, setting clear goals, following best practices, striving for operational excellence, or exceeding benchmarks.	performance expectations set for employees, goal setting routines, accountability for targets, productivity practices, operational excellence routines, best-practice adoption, benchmarking of units or teams, incentive links to performance.	revenue growth, record earnings, margins, guidance or market share reported as outcomes; these are results, not expectations governing conduct.

Category	Definition	Indicators	Non-qualifying signals
Other reported norms (residual; adapts Li et al.'s Miscellaneous)	Reported expectations, rules or routines concerning conduct that does not fit the five categories above. Generic culture declarations without reported conduct are NOT coded here; they are counted in the declaration ledger.	reported rules or routines about conduct that no specific category covers, e.g. "every manager must hold a weekly one-to-one", when no category indicator applies.	generic references to "our culture" or "cultural change" without reported conduct (ledger); any item that fits a specific category above; any item with no normative content.

Repeated conduct. A recurring pattern is retained as a descriptive subtype (`content_type = routine`) and supports a normative interpretation only when the record states it as standard practice for a defined set of actors. Repetition inferred by the agent, or an isolated episode, caps relevance at weak unless an expectation, rule or sanction is also reported. Routines are therefore reported practices treated as evidence of expectations, and the final record keeps the content type so that users can separate stated expectations from reported routines.

B.4 Judge prompt

```
You are the Judge. You receive the interpretation view of the evidence record,
the codebook (six category definitions), and six independent category
assessments. You do not receive the transcript or the declaration ledger.

TASK: propose CLAIMS for this passage. A claim is one expectation, rule or
routine assigned to one category; a category may carry several claims and a
conduct item may support claims in more than one category. For each claim
give: category; status (supported | partially supported | insufficient
evidence); evidence IDs; the reported conduct; content_type; direction;
origin; scope_type, scope_target, scope_extent; conduct status (completed |
ongoing | planned). Categories with no claim are "insufficient evidence".

RULES
1. Every claim must cite evidence IDs from the view. Nothing else counts.
2. Plans keep prospective status; they never support implemented practice.
3. An episode supports a norm only if the view states a rule, routine or
   repeated conduct. Say explicitly when you generalize from an event.
4. Outcomes (growth, records, margins) are not conduct expectations.
5. Copy direction, origin and scope fields from the view; do not widen them.
6. If completeness_uncertain is true, say which claims it could affect.
7. State the strongest objection to your own claims in one sentence.
Output valid JSON (judgment schema).
```

B.5 Interpretive critic prompt

```
You are the Interpretive Critic. You receive the same interpretation view,
codebook and six assessments as the Judge, and the Judge's claims. You do
not receive the transcript.

TASK: test the link from evidence to construct for each supported or partially
supported claim. Ask, in order: (a) is there an alternative reading under
which the conduct is not a normative expectation at all (a one-off decision,
an outcome, a plan)? (b) does the claim generalize from an episode to a norm
```

```
without a stated rule or routine? (c) does it infer shared culture from a
formal rule whose adoption is unobserved? (d) does it rely on a plan or
declaration? (e) are scope_type, scope_target or scope_extent wider than the
view states? (f) are direction, origin or content_type misstated? An
externally imposed requirement (regulation, customer) is still reported
normative content: it changes the origin field and the stated limits, not the
category. (g) does an unresolved source objection undercut the claim?

For each objection cite evidence IDs and the coding criterion. Propose the
narrowest claim the view supports. If no objection is warranted, accept the
claim and say why. Do not add new evidence. Output valid JSON (critique schema).
```

Round two. The Judge replies to the challenge, accepting, narrowing or defending each claim with evidence IDs. The critic then reassesses the revised claims. In arm E (Section D.2) the Judge re-reads its own claims without the challenge and the critic reassesses the original claims without the reply, so that each side still produces two statements.

B.6 Adjudicator prompt

```
You are the Adjudicator. You receive the interpretation view of the evidence
record, the codebook, and four statements: Judge claims, critic challenge,
Judge reply, critic reassessment. You do not receive the transcript, any
identifiers, or any data about the firm's outcomes.

TASK: produce the FINAL MEASUREMENT RECORD for this passage: for each category,
a list of claims (possibly empty), each with relevance (none | weak | moderate
| strong); the reported conduct; content_type; direction; origin; scope_type,
scope_target, scope_extent; conduct status; evidence IDs for and against; and
limitations, each tagged: SOURCE_INHERENT only if the source stage marked the
missing information as absent from the passage (source_states = absent) or
the limitation is of a kind no passage could settle (e.g. compliance with a
stated rule); RESOLVABLE if a human reading the passage could settle it;
UNDETERMINED if you cannot tell whether the gap is in the source or in the
record. Also give the category-level summary: relevance = the maximum over
its claims; direction = the direction of the strongest claim, or "mixed" when
claims of equal strength disagree.
Separately, record the PROCESS: your action per claim (accept | narrow |
withhold relative to the Judge's claims) and the objections left unresolved.

RULES
1. Decide on evidence IDs, not on how persuasive a statement sounds.
2. Agreement between Judge and critic is not proof; check the evidence yourself.
3. If an objection is unanswered by evidence, record it as unresolved and
   narrow or withhold accordingly. A forced label is never required.
4. Set escalate_to_human = true if any RESOLVABLE or UNDETERMINED limitation
   remains, or if completeness_uncertain is true.
5. Apply the codebook as written. Do not reinterpret definitions.
Output valid JSON (measurement schema and process schema).
```

B.7 Output schemas

All agents emit JSON validated against the schemas in the repository (`schemas/`). Abbreviated forms:

```
record:        {passage_id, items:[{id, actor, conduct, timing, status, content_type,
                direction, origin, scope:{type, target, extent}, qualifications:[...],
                excerpt, excerpt_span}], declarations:[...],
                unresolved_flags:[{type, target_item, source_states, note}]}
record_interp: {passage_id, items:[{id, actor, conduct, timing, status, content_type,
```

```
                direction, origin, scope:{type, target, extent}, qualifications:[...],
                excerpt}], unresolved:[{type, evidence_ids, source_states, note}],
                completeness_uncertain: bool}
                (derived view: no declarations, no spans, no passage text)
critique:      {passage_id, flags:[{id, type: UNSUPPORTED_CLAIM | OMITTED_QUALIFIER
                | IGNORED_COUNTEREVIDENCE | SCOPE_OR_STATUS_ERROR | WEAK_EXCERPT
                | MISSING_ITEM, target_item, sentences, question}],
                complete_and_supported: bool}
assessment:    {passage_id, category, entries:[{evidence_ids, relevance, direction,
                origin, scope:{type, target, extent}, reasons_for:[...],
                reasons_against:[...], unresolved:[...]}]}
judgment:      {passage_id, claims:[{category, status, evidence_ids, conduct,
                content_type, direction, origin, scope:{type, target, extent},
                conduct_status, generalized_from_event: bool}], strongest_objection}
measurement:   {passage_id, categories:{<cat>:{relevance, direction,
                claims:[{relevance, conduct, content_type, direction, origin,
                scope:{type, target, extent}, conduct_status, evidence_for:[...],
                evidence_against:[...], limitations:[{text, kind:
                SOURCE_INHERENT | RESOLVABLE | UNDETERMINED}]}]}}}
process:       {passage_id, actions:[{claim_ref, action: accept | narrow | withhold}],
                unresolved_objections:[...], escalate_to_human: bool,
                output_state: valid | abstained | invalid}
```

Table B3. Which field of the final measurement or process record each metric scores.

Metric (Section D.5)	Field scored	Reference
Category relevance agreement	measurement.categories.<cat>.relevance (category summary)	Adjudicated expert relevance (Layer 2)
Direction agreement; correct change on normative-flip edits	measurement.categories.<cat>.direction; claim-level direction for the edited claim	Expert direction; frozen expected change
Scope agreement; correct change on scope edits	claims[].scope.type, .target, .extent (all three must match)	Expert structured scope; frozen expected change
Correct invariance on irrelevant edits	Category relevance, direction and scope of every claim, jointly	Unchanged and equal to the expert reference on both members of the pair
Evidence completeness (C–D and secondary)	record.items vs. reference conduct items	Expert Layer 1 labels
Decision coverage	process.output_state = valid for the passage-category pair (relevance none counts as a decision)	Not reference-based
Erroneous exclusion	relevance none or abstained where reference relevance is weak or above	Expert Layer 2
Escalation rate (workload)	process.escalate_to_human	Not reference-based
Operational failure rate	process.output_state = invalid after retry	Not reference-based

C. Worked example

The passage, record and statements below are constructed to show the information flow. They are not model output; recorded runs are in the repository logs.

Masked passage. "We champion innovation. We stopped twelve pilots this year; teams must share unsuccessful results at monthly reviews."

Reader record (initial). E1: actor [COMPANY_001]; conduct: stopped twelve pilots; timing: this year; status: completed; content_type: episode; direction: unclear; origin: unstated; scope: organization-wide / [COMPANY_001] / all; qualifications: none; excerpt "We stopped twelve pilots this year". Declaration ledger: "We champion innovation".

Skeptic critique. F1 (MISSING_ITEM, sentence 2): the second clause reports a requirement, "teams must share unsuccessful results at monthly reviews"; it appears nowhere in the record. F2 (SCOPE_OR_STATUS_ERROR, E1): the passage does not say which units stopped pilots; is organization-wide stated or inferred?

Reader reply and revised record. E1 scope corrected to unspecified / none / unspecified, source_states = absent. E2 added: actor "teams"; conduct: share unsuccessful results at monthly reviews; timing: ongoing; status: ongoing; content_type: routine (stated as standard practice, "must ... at monthly reviews"); direction: required; origin: internal; scope: team or role / teams / all (which teams: absent); qualifications: compliance not reported; excerpt "teams must share unsuccessful results at monthly reviews". Unresolved: which teams (source_states = absent). completeness_uncertain = false.

Six assessments (interpretation view only). Innovation and adaptability: one entry, relevance strong, direction required, scope team or role / teams / all; reasons_for E2 (a stated routine for learning from failed experiments), E1 (experimentation implied by pilots, episode); reasons_against: compliance unobserved. Collaboration and people-focused: relevance weak; E2 involves sharing within teams but no conduct about cooperation or well-being. Performance-oriented, customer-oriented, integrity and risk management, other reported norms: relevance none.

Judge claims. Claim 1: innovation and adaptability; supported; evidence E1, E2; conduct: sharing unsuccessful results at monthly reviews; content_type routine; direction required; origin internal; scope organization-wide / [COMPANY_001] / all; status ongoing. Strongest objection: the view does not say which teams are covered.

Critic challenge. Objection (e): the view gives scope team or role / teams / all, not organization-wide; the Judge widened it. Objection (c): E2 is a requirement; adoption is unobserved. Objection (b): E1 is an episode; it supports experimentation only together with E2. Narrowest supported claim: a required learning routine for teams covered by the monthly review.

Judge reply. Accept (e) and (c): scope restored to team or role / teams / all; limitation "compliance unobserved" added. On (b): E1 retained as context, not as independent support.

Critic reassessment. No further objection. The narrowed claim is supported by E2 with E1 as context; the compliance limitation must appear in the record.

Final measurement record. Innovation and adaptability: category relevance strong, direction required; claim 1: relevance strong; conduct: teams share unsuccessful results at monthly reviews;

content_type routine; direction required; origin internal; scope team or role / teams / all; status ongoing as reported; evidence_for E2 (E1 context); evidence_against none; limitations: compliance unobserved (SOURCE_INHERENT, no passage could settle it); which teams (SOURCE_INHERENT, source_states = absent). All other categories: relevance none. Process: claim 1 narrow; unresolved objections none; escalate_to_human false; output_state valid. The declaration "We champion innovation" was never available to the interpretation stage.

Profile entry. Firm [COMPANY_001], quarter Q: innovation and adaptability, 1 claim (required routine), supporting evidence 1 (E2), contradictory 0, source-inherent limitations 2; declaration ledger count 1; coverage: 1 passage; source link to passage_id.

D. Validation protocol

D.1 Paired edits (behavioral tests)

We adapt CheckList-style behavioral testing (Ribeiro et al., 2020) to pairs consisting of an original passage and an edited version. Edit templates and instructions are developed on development firms; the evaluation pairs are constructed from held-out test-firm passages, checked by two annotators for edit type and for the absence of any other change, and the warranted response to every pair is frozen by the domain experts on the team, who do not write prompts, in the preregistration before any model output is seen. Development-firm pairs may be shown as illustrations and are never counted as confirmatory results.

- **Irrelevant, promotional addition.** Insert a value declaration or self-congratulatory sentence with no reported conduct ("We are proud of our world-class culture of innovation"). Expected: relevance, direction and scope unchanged.
- **Irrelevant, equivalent paraphrase.** Reword reported conduct without changing actor, conduct, timing, status, content type, direction or scope. Expected: unchanged.
- **Relevant, normative flip.** Hold the conduct and sentence structure constant and change only its direction: "teams are encouraged to share unsuccessful results at monthly reviews" versus "teams are discouraged from sharing unsuccessful results at monthly reviews". Expected: the claim's direction changes from encouraged to discouraged; relevance and scope unchanged.
- **Relevant, scope change.** Change who the conduct covers while holding type constant: "one plant must conduct the review" (unit or function / plant / one) versus "every plant must conduct the review" (unit or function / plants / all). Expected: scope_extent changes as specified; direction unchanged.

Scoring. A pair is counted as correct only if the prespecified field changes in the prespecified direction (relevant edits) or every scored field stays unchanged (irrelevant edits), and the responses to both the original and the edited passage match the expert reference. "Any change" is not success, and "no change" is not success if both responses are wrong. Target set size: 200 pairs, half irrelevant and half relevant, all from test firms.

D.2 Arms A to F

Table D1. Arm definitions. In every row, everything not listed is identical across the two arms.

Arm	What differs	Budget matching
A (treatment)	Six experts and Judge receive the interpretation view only.	Same model, same prompts except for the presence of the passage, same generation limits as B.
B (control)	Six experts and Judge additionally receive the full masked passage and context window. Record, critic, adjudicator and all exchanges unchanged.	Input tokens differ by construction; output limits equal. Both reported.
C (control)	After the skeptic critique is generated, it is withheld. The Reader re-reads the passage with a generic self-check instruction covering the same six error types and produces a revised record.	One Reader call in both arms; the self-check prompt has the same length band as a critique.
D (treatment)	The Reader receives the same skeptic critique and replies to it, producing the revised record.	As C.
E (control)	Round two without exchange: Judge re-reads its own claims and may revise; critic reassesses the original claims. Neither sees the other's second statement.	Four statements, same JSON format and length limits as F.
F (treatment)	Round two with exchange: challenge passed to the Judge; reply passed to the critic.	As E.

Table D2. Outputs shared across the two arms of each contrast, and outputs regenerated within each arm. Anything affected by the intervention is regenerated.

Contrast	Shared (generated once)	Regenerated in each arm
A–B: narrative access	Reader record, skeptic critique, Reader reply, revised record and interpretation view	Six assessments, Judge claims, critic challenge, Judge reply, critic reassessment, adjudication
C–D: access to the critique	Initial Reader record, skeptic critique	Reader revision (reply or self-check), revised record, interpretation view, and all interpretation-stage outputs through adjudication
E–F: interpretive exchange	Revised record, interpretation view, six assessments, Judge claims, first critic challenge	Judge reply, critic reassessment, adjudication

Budget matching means matched call counts and generation limits per role, not identical total computation: input lengths differ by construction in A–B. For each arm we report input tokens, output tokens, wall-clock latency and cost at the provider's list price on the run date, per passage and in total. The adjudicator prompt and model are identical across all six arms. Two kinds of repetition are reported and kept distinct: (i) reproducibility under fixed decoding, by re-running each arm three times at

temperature 0 with the pinned runtime, where differences reflect serving nondeterminism only; and (ii) stochastic robustness, by re-running each arm on a subset of 100 passages at temperature 0.7 with three seeds, reported as a sensitivity analysis and not as the primary estimate.

D.3 Human-swap diagnostics

On a prespecified subset of 100 validation passages, one agent's output is replaced by an independently produced human counterpart (a human-written record, a human-written critique, or human claims). "Holding other agents fixed" means holding their models and prompts fixed and regenerating every downstream output from the substituted input. For each swap we report whether the final measurement record, compared with the expert reference, was corrected (wrong before, right after), broken (right before, wrong after) or unchanged in substance. The purpose is to locate where errors enter, not to decompose error exactly; swaps are not independent of one another.

D.4 Annotation protocol and expert reference labels

Sampling: approximately 500 passages, 400 from at least 40 test firms and 100 from at least 10 development firms, sampled before any screening and stratified by evidence type (completed action, ongoing practice, plan, declaration only, mixed), with about ten passages per firm and at least 40 per expected category. Annotators: coders with graduate training in management or finance research, paid at the institution's standard research-assistant rate; they never see model outputs. Procedure: (1) calibration on a separate sample of 50 development-firm passages, during which the instructions are refined; (2) the codebook and instructions are frozen and preregistered, together with any decision to exclude a field from the primary outcomes on reliability grounds, which must rest on calibration evidence only; (3) every validation passage is coded independently by two annotators; (4) disagreements are adjudicated by a third senior coder; (5) the adjudicated labels are the expert reference. Reliability (Krippendorff's alpha) is reported per field and per category on the independent codes before adjudication; it guides training and is not treated as proof of validity. The central fields (relevance, direction, structured scope) are never dropped after seeing test data: if calibration shows that one of them cannot be coded reliably, that limits the conclusions drawn for it and is stated as such. Two layers are coded, by different people where possible:

- **Layer 1, source.** For each reported conduct item: supported by the passage; qualifiers omitted; conduct items missing; timing, status, content type, scope fields and direction correct; and, for each gap, whether the passage states the information (absent, unclear, present).
- **Layer 2, interpretation.** For each category: relevance (none, weak, moderate, strong); for each claim, normative direction, origin, structured scope (type on the controlled vocabulary; target as a short string normalized to the placeholder or unit name; extent as one, some, all, named or unspecified), conduct status, and content type (expectation, rule, routine, episode). Scope is scored field by field, so equivalent prose descriptions cannot be penalized.

D.5 Metrics, inference, preregistration and comparators

- **Primary outcomes (preregistered).** (i) Agreement with the expert reference per field: category relevance (macro-F1 over categories and exact agreement), claim direction and structured scope (exact agreement on all scope fields); (ii) correct invariance rate on irrelevant edits; (iii) correct change rate on relevant edits, both scored as in D.1.

- **Two coverage quantities, kept distinct.** *Evidence completeness* is the share of reference conduct items present in the final record; it measures the source stage and is the main secondary outcome for C–D. *Decision coverage* is the share of passage-category pairs for which the system returns a valid substantive decision (including a valid relevance none) rather than an abstention or an invalid output; it measures the interpretation stage. Because A and B share the same revised record, only decision coverage can detect additional withholding by the interpretation stage.
- **Withholding safeguard.** For A–B, the preregistered success criterion is a reduction in false changes on promotional edits with relevant-edit sensitivity and decision coverage each no more than five percentage points below arm B (one-sided 95 percent bound); the tolerance is fixed in the preregistration. An improvement obtained by withholding is a failure. Error rates are also compared at matched decision coverage by counting a decision as substantive only at or above a relevance cut-off (strong; moderate or above; weak or above) applied identically to both arms.
- **Output states.** Every passage-category pair ends in one of three states: a valid substantive response (including a valid none), an explicit abstention (withhold with insufficient evidence), or an invalid response after the one permitted retry. Invalid responses fail the primary correctness endpoints, remain in their denominators, and feed a separately reported operational failure rate; they never receive credit when the reference is none.
- **Other secondary outcomes.** Erroneous exclusions (reference-relevant pairs decided as none or abstained), escalation rate as human-review workload, run-to-run agreement, and Layer 1 agreement for the C–D contrast.
- **Inference.** Arm differences are estimated on test firms. Uncertainty for all statistics, including macro-F1, uses paired resampling of firms, keeping every passage, edit pair, arm and repeated run within a sampled firm together, so that firm-level dependence and the paired design are preserved. Minimum margins of five F1 points for relevance agreement and five percentage points for edit correctness are preregistered.
- **Preregistration.** Prompts, codebook, edit templates and expectations, margins, reliability-based exclusions and analysis code are registered on OSF before any test-firm passage is processed by the final pipeline; the registration link is added to the repository README at that point. Development firms are used freely before that date and never for reported results.
- **Comparator 1, single-pass baseline (the architectural comparison).** One prompt receives the raw masked passage and the identical codebook, target construct (reported normative content), admissible-evidence rules and output schema as the full instrument, and returns the same measurement fields in one pass with the same pinned model. It is scored against the same reference. Li et al. (2026, footnote 5) also report a single-pass variant of their own pipeline, which supports a single-pass comparator; ours differs in using our codebook and target so that the comparison isolates architecture. This is the simple baseline; its performance is an empirical question.
- **Comparator 2, faithful implementation of the published literature method.** Li et al. (2026) apply their generative pipeline to earnings calls as well as analyst reports, with modifications described in their Internet Appendix; for earnings-call components the retrieval-augmentation step used for analyst reports is not applied (Internet Appendix A.7). We implement their

earnings-call procedure with their released prompts where the replication package provides them, documenting any adaptation, and with their original Miscellaneous definition, on the same passages and pinned model, and report it separately from Comparator 1, stating the differences in target (their culture-type classification versus our reported-norm claims) and in admissible evidence. Any correlation of profiles with firm outcomes is supplementary and is not an evaluation criterion.

E. Reproducibility and compute

Primary model: a pinned open-weight model whose name, version and weights checksum are recorded in the repository configuration, run with temperature 0, a fixed seed and a fixed maximum output length per call, served by a pinned runtime version. Second family for robustness: a second open-weight model family, recorded the same way. All prompts are versioned in the repository and every call is logged with the prompt hash, inputs (including the interpretation view actually passed), outputs, token counts and latency. Logs for the validation set are released in full. Known limits: deterministic decoding does not guarantee identical outputs across hardware or runtime versions, so we report the runtime version and the re-run agreement rate from D.2.

Tokens in and out, latency and cost are reported per passage for the full design and for both comparators, per firm-quarter, for the validation set under all six arms and repetitions, and extrapolated to a full cross-section; the figures are written to a cost log in the repository with every run and summarized in the article.

F. Ethics and data access

- **Transcripts.** Redistribution follows the provider's terms. The replication subset is built from publicly filed transcripts, and source text inside released logs is limited to the verified excerpts of at most 25 words each, so that logs can be released even where full transcripts cannot.
- **Annotators.** Recruited through the authors' institution with informed consent and paid at its standard research-assistant rate. Ethics approval from the institution's research ethics committee is obtained before annotation begins; the reference is withheld for anonymous review.
- **Model terms.** Open-weight models are used under their published licenses; any proprietary model used for robustness is named with its terms of use, and no result in the article depends on it alone.
- **Subjects of measurement.** Profiles describe executives' reported accounts at the firm level. No individual employee is identified or characterized. Documentation released with the instrument states that profiles are not assessments of members of the firm and must not be used as such.
- **Re-identification.** The probe in A.3 is reported; masked passages in the public release are additionally screened by hand for residual identifiers.
- **Misuse.** The instrument produces evidence-linked claims with explicit limitations; stripping those fields to produce a bare score defeats its purpose, and the release documentation says so.

G. Repository

The anonymized repository accompanying this submission carries no author-identifying metadata in commits, paths or README. Contents:

- `prompts/`: the prompts of Section B, the earlier prototype prompts kept for comparison, and the codebook (Section B.3).
- `schemas/`: JSON schemas for every object in Section B.7, including `record_interp`, `measurement` and `process`.
- `pipeline/`: the five-role prototype, which runs end to end, and the extension to the full design under development, with arm switches for A to F and the view constructor that builds `record_interp`.
- `demo/`: a small set of passages from publicly filed transcripts plus the constructed passage of Section C, and `make demo`, which runs the pipeline on them with the pinned model and writes new logs. The constructed exchange in Section C is explanatory; the demo produces real model output on the same input.
- `logs/`: JSON logs of the prototype pilot runs described in Section 5 of the proposal and summarized in Section H, including the inputs actually passed to each call, and the pilot summary (passages processed, flag and challenge rates, narrowing and withholding, escalations, tokens and cost per passage).
- `validation/`: edit templates, annotation guidelines and controlled vocabularies, preregistration draft, analysis code including the firm-level resampling.
- `README.md`: installation, model download and checksum, expected runtime per passage, and the exact demo command.

H. Preliminary results from the prototype pilot

This section reports the first runs of the five-role prototype. The figures are descriptive, never validation: there is no expert reference yet, the passages were not chosen for cultural content, and each figure comes from a single run at fixed decoding. All figures are computed from the released logs (Section G), which hold every call's rendered prompt, exact inputs, raw replies, validated output, token counts and latency; the repository's pilot summary reproduces Table H1.

H.1 Set-up

Model: Qwen2.5-14B-Instruct, an open-weight model released under the Apache-2.0 license, in 4-bit quantization (Q4_K_M), served by Ollama 0.40.13 with a 20,480-token context on one consumer GPU with 16 GB of memory, at temperature 0 with a fixed seed; the weights digest is recorded in the repository configuration. Prompts and checks: the prompts of Section B, the validators behind Section B.7 and one retry per call after validation feedback. Configuration: the full design (arms A, D and F) without the optional expertise stage, eight calls per passage. Passages: the constructed passage of Section C; four passages selected by the authors from an earnings-call transcript filed with the regulator as an exhibit to a current report, a public record, covering prepared remarks of two executives and two question-and-answer exchanges; and twelve masked windows taken by position, four from each of three licensed transcripts of calls held on 30 January 2026. Released logs of the licensed windows replace the source text with a placeholder and keep only the verified quotes (Section F).

Table H1. Prototype pilot, full design, five roles, 17 passages. Counts refer to the 16 passages that completed unless stated; figures are descriptive.

Quantity	Value
Passages processed	17: 12 valid, 4 explicit abstentions, 1 invalid after the permitted retry
By source (valid / abstained / invalid)	constructed 1 / 0 / 0; public 3 / 1 / 0; licensed 8 / 3 / 1
Reader's conduct items, initial and revised	43 and 62 (16 passages)
Declarations in the ledger	11
Skeptic flags	52 in 16 of 16 passages: omitted qualifier 31, missing item 20, ignored counterevidence 1
Reader responses to flags	item added 21, defended 19, amended 12
Unresolved source objections in the view	36; completeness uncertain in 6 of 16 passages
Judge claims, round 1	28: supported 5, partially supported 2, insufficient evidence 21
Critic objections, round 1	23: episode generalized to a norm 15, alternative reading 7, culture inferred from a formal rule 1; 5 claims accepted without objection
Judge responses to objections, round 2	objection accepted 18, claim narrowed 3, claim defended 2
Judge claims, round 2	12: supported 4, partially supported 2, insufficient evidence 6; no claim left in 9 passages
Critic objections, round 2	22; 1 round-1 objection resolved
Adjudication of round-2 claims	accepted 1, narrowed 2, withheld 9 (6 of the withheld already marked insufficient evidence by the Judge)
Claims in the final measurement records	3 in 3 passages: performance-oriented, strong, required routine (2); collaboration and people-focused, weak, required routine (1)
Limitations on retained claims	source-inherent 2, resolvable 1
Passages escalated to human review	6 of 16
Boundary checks on interpretation calls; violations	80; 0
Calls needing the one permitted retry	43 of 129: adjudicator 13, critic round 2 9, Judge round 1 8, Reader (revision) 7, Reader (initial) 4, Judge round 2 2
Invalid passages	1: licensed passage C096470cf42:w003, Reader (initial): excerpt over the 25-word limit after the retry
Tokens per completed passage, mean	36,841 input and 4,304 output with retries (27,042 and 3,028 without); local inference, no marginal price
Model time per completed passage	mean 63.0 s, median 61.1 s, retries included

H.2 The demonstration passages

The recorded run on the passage of Section C (repository, demo/recorded) differs from the explanatory exchange there. The Reader recorded the review rule in its first pass (teams; share unsuccessful results at monthly reviews; routine; required; internal; scope unit or function, teams, all) and placed "We champion innovation" in the declaration ledger. The skeptic's only flag asked about an omitted qualifier, which the Reader defended. The Judge assigned the rule to collaboration and people-focused, reading it as information sharing among teams rather than as experimentation, and the critic objected that the claim inferred shared culture from a formal rule whose adoption is unobserved (criterion c). The Judge narrowed the claim, and the adjudicator recorded collaboration and people-focused, weak relevance, a required internal routine for the teams covered, with a source-inherent limitation that adoption is unobserved and no escalation; the declaration never reached the interpretation stage. Which category, or categories, the rule warrants is the kind of construct question the expert reference is designed to settle; Section C illustrates the information flow, not the instrument's output.

The four publicly filed passages, run in the same batch, show the other outcomes. On the chief executive's remarks about efficiency and revenue per employee, the Judge proposed four performance-oriented claims and dropped the two that the critic challenged as episodes generalized to norms; of the remaining two, the adjudicator withheld one and kept a narrowed claim that the stated focus on efficiency is a required organization-wide routine, with a resolvable limitation (a generalization from an episode to a norm without a stated rule) and escalation to human review. On the chief financial officer's answer about where investment spending appears, the critic objected that the passage explains past spending rather than stating an expectation; the Judge withdrew its claim and the record contains no culture claim. On the milestones since 2020 (recurring-revenue conversion, debt repayment, cost reduction and margin growth), the Judge defended a performance-oriented claim as a pattern of conduct, the critic objected that episodes do not establish a norm without a stated rule, and the adjudicator withheld the claim, leaving no culture claim. On the remarks about supplier-data accuracy, the Reader added five conduct items in response to the skeptic, the Judge withdrew its integrity claim after the critic's objection, and the adjudicator abstained and escalated because the record's completeness remained uncertain.

H.3 Control arms and the expertise stage

Each control arm has been run once on the constructed passage through the same controller, reusing the shared calls of Table D2 from the cache: B (experts and Judge also receive the passage), C (critique withheld, generic self-check instead) and E (round two without exchange). All three produced valid records containing the same narrowed collaboration and people-focused claim as the full design. The fourteen-call configuration with the six expertise agents has run once on the same passage: the agents rated five of the six categories as relevant to some degree, the Judge proposed claims in five categories, and the adjudicator withheld all five and abstained, noting that a requirement to share results does not by itself establish a collaborative environment. These runs show that the switches and the optional stage execute end to end; they are not comparisons, which require the reference labels and the sample of Section D.

H.4 Predecessor pilot

The predecessor pilot (May 2026) simulated ten roles in a single prompt on a hosted model (six category specialists, tone, causal structure, an evidence verifier and a judge) and applied them to the

screened executive components of 17 earnings calls held on 29 and 30 January 2026; 16 replies were parseable. In 15 of the 16 calls two or more specialists recommended a category, yet the judge returned one type for every call (innovation and adaptability 11, performance-oriented 4, integrity and risk management 1) and the tone role returned positive for all 16; the judge's type matched the highest-confidence recommendation in 7 calls. The verifier rated 15 of 16 outputs highly faithful while checking quotations against the same passage the other roles had read. Tokens and cost were not logged. The prompt, the extraction scripts and the summary figures are in the repository (legacy_pilot/); the per-call outputs are withheld because they quote the licensed transcripts at length.

H.5 What the pilot shows and does not show

The pilot shows that a 14-billion-parameter open-weight model on one consumer GPU can meet the output contracts of Section B for 16 of 17 passages, with one retry in a third of calls (43 of 129); that every boundary check on interpretation-stage inputs passed (80 calls); and that the source stage changes the record, with 21 items added in response to the skeptic's questions. It does not show that any judgment is correct. It also surfaces two behaviors for the development phase: the Judge conceded 18 of 23 objections, and only three claims reached the final records. Both are consistent with passages that mostly report results rather than conduct, but they could also reflect the over-conservative critic named among the key risks in the proposal; the decision-coverage tolerance and the erroneous-exclusion rate (Section D.5) are designed to tell these apart.

References cited in this appendix

Li K, Mai F, Shen R, Yang C, Zhang T (2026) Dissecting corporate culture using generative AI. *Review of Financial Studies* 39(1):253–296.

Ribeiro MT, Wu T, Guestrin C, Singh S (2020) Beyond accuracy: Behavioral testing of NLP models with CheckList. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 4902–4912.